

Big Data Logistics & Supply Chain Analytics Dashboard

Internship Project Report

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Organization

**Zaalima Development Pvt.
Ltd**

Department

Data Analytics

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Project Overview

The Logistics and Supply Chain industry generates massive amounts of operational data every day. This project utilizes Big Data Analytics techniques to process, analyze, and visualize logistics data using PySpark and Power BI.

The goal is to improve operational efficiency, monitor inventory, evaluate supplier performance, and support business decision-making through interactive dashboards.



Big Data Analytics

PySpark-powered processing



Inventory Monitoring

Real-time stock tracking



Logistics Efficiency

Operational optimization



Business Insights

Data-driven decisions

Problem Statement & Objectives

⚠ Challenges

- Delivery delays
- High transportation costs
- Inventory management issues
- Supplier performance monitoring
- Lack of data-driven decisions

✓ Objectives

- Analyze logistics performance
- Monitor inventory levels
- Evaluate supplier efficiency
- Forecast future demand
- Develop interactive dashboards

Technologies Used



Python

Data Analysis



PySpark

Big Data Processing



SQL

Database Queries



Pandas

Data Manipulation



Power BI

Dashboard Development



Google Colab

Development Platform

Project Architecture



This architecture enables efficient processing of large logistics datasets and transforms raw data into actionable insights.

Data Processing & PySpark Workflow

Data Preparation

- Missing value handling
- Duplicate removal
- Data validation
- Feature engineering

PySpark Operations

- Data ingestion
- Data transformation
- Aggregation & grouping
- KPI generation
- Export for visualization

Benefits

- Faster processing
- High scalability
- Handles large datasets

Key Analysis Performed

Revenue Analysis

- Total revenue trends
- Product-wise revenue

Transportation

- Shipping cost analysis
- Delivery performance

Inventory Analysis

- Stock monitoring
- Inventory turnover

★ Supplier Performance

- Rating comparison
- Delivery efficiency

Demand Forecasting

- Future demand prediction
- Trend analysis

Power BI Dashboard & KPIs

Dashboard Features

- Interactive filters
- Drill-down analysis
- Dynamic visualizations
- Real-time KPI monitoring

Key Performance Indicators



Total Revenue



Products Sold



Avg Shipping Cost



Avg Stock Level



Supplier Score



Demand Forecast

Results & Learning Outcomes

Project Results

- Improved supply chain visibility
- Better inventory planning
- Identification of high-cost routes
- Enhanced supplier evaluation
- Support for strategic decisions

Learning Outcomes

- PySpark Data Processing
- Power BI Dashboard Development
- Business Intelligence Concepts
- Data Analytics Workflow
- Real-world Project Experience

Conclusion

This project demonstrates how Big Data Analytics and Business Intelligence tools can optimize logistics and supply chain operations. By leveraging PySpark and Power BI, the solution provides valuable insights that improve efficiency, reduce costs, and support data-driven business decisions.

Thank You

Questions & Discussion